ReDI SCHOOL

Objects & Final Project

Week 9

Agenda

1. Homework Recap
2. Recap
   a. Arrays
   b. Loops
3. Objects
4. Introduction to Final Project
5. Ideation Tips for Your Final Project



Homework Recap



Homework Discussion - 10min



Recap - Arrays

How to get elements from an array



You can access and retrieve elements from an array using its index.
You need to use the square bracket syntax to access array elements.

```
const element = array[index];
```

When you're accessing elements using index it looks like this:

```
const salad = ['🍅', '🍄', '🥦', '🥬', '🌽', '🥕', '🥑'];
salad[0]; // '🍅'
salad[2]; // '🥦'
salad[5]; // '🥕'
```

```
const info = ['Katherine', 'Johnson', '101'];
const firstName = info[0]
const lastName = info[1]
const age = info[2]
```



Recap - Arrays

Array.isArray()

```
Array.isArray([27, false, 'ReDI School', ,(a,b)=>a+b, {}]); // returns true
Array.isArray('🍅'); // returns false
Array.isArray({ 'tomato': '🍅'}); // returns false
Array.isArray([]); // returns true
```



Recap - Arrays

Array Iterator Methods: filter()



The filter() method creates a new array with all the elements that satisfies the condition mentioned in the function. Let's find the students whose name includes an **r**.

```
44
45    const studentsWithR = students.filter((element, index) => {
46        return element.includes('r')
47    })
48
```

```
34
35    let students = ['Alex', 'Javi', 'Maria', 'Jack', 'Ibra', 'Rhona']
36
```



Recap - Arrays

Array Iterator Methods: map()



The map() method creates a new array by iterating through the elements and applying logic we provided in the function as an argument. We will make all of the students names be in capital letters.

```
51
52    const studentsUppercase = students.map((element, index) => {
53        return element.toUpperCase()
54    })
55
```



Recap - Arrays

```
34
35    let students = ['Alex', 'Javi', 'Maria', 'Jack', 'Ibra', 'Rhona']
36
```

```
56
57    const totalNumberOfLetters = students.reduce(
58        (accumulator, student) => {
59            accumulator = accumulator + student.length
60            return accumulator
61    }, 0)
62
```

- We initialize the accumulator with 0.
- We apply the reduce method on each of the student names. We retrieve the name length and add it to the accumulator.
- Finally, we return the accumulator.



Recap - For loop

Loops are handy, if you want to run the same code over and over again, each time with a different value. Let's take a look at the first loop type:

```
for (initialization; condition; finalExpression) {
  // the code to run
}
```

initialization - This expression runs before the execution of the first loop, and is usually used to create a counter.

condition - This expression is checked each time before the loop runs. If it evaluates to true, the statement or code in the loop is executed. If it evaluates to false, the loop stops. And if this expression is omitted, it automatically evaluates to true.

finalExpression - This expression is executed after each iteration of the loop. This is usually used to increment a counter, but can be used to decrement a counter instead.



Recap - For loop over an array

There are two ways to fix this code: set condition to either i < arr.length or i <= arr.length - 1.

```
34
35    let students = ['Alex', 'Javi', 'Maria', 'Jack', 'Ibra', 'Rhona']
36
```

```
for (let i = 0; i < arr.length; i++) {
  console.log(arr[i]);
}
```

```
for (let i = 0; i <= arr.length - 1; i++) {
  console.log(arr[i]);
}
```



Recap - For-of loop

The JavaScript **for-of** statement loops through the values of an iterable object.

It lets you loop over iterable data structures such as Arrays, Strings, Maps, NodeLists, and more:

```
const listContainer = document.getElementById('List-container');
const carBrands = ['BMW', 'Volvo', 'Saab', 'Ford', 'Fiat', 'Audi'];

for (let x of carBrands) {
  const node = document.createElement('li');
  node.innerHTML = x;
  listContainer.appendChild(node);
}
```



Recap - For-in loop

The JavaScript for in statement loops through the properties of an Object:

- The for in loop iterates over a person object
- Each iteration returns a key (x)
- The key is used to access the value of the key
- The value of the key is person[x]

```
const person = {
  firstName: 'John',
  lastName: 'Doe',
  age: 25,
};

let text = '';
for (let x in person) {
  text += person[x];
}
```



Objects

An object is a collection of properties, and this properties consist of a name (or key) and a value pair.

Objects in JavaScript, just as in many other programming languages, can be compared to objects in real life.

Compare it with a cup, for example. A cup is an object, with properties. A cup has a color, a design, weight, a material it is made of, etc. The same way, JavaScript objects can have properties, which define their characteristics.

```
const person = {
  firstName: 'Emilia',
  lastName: 'Doe',
  age: 55,
  hasSiblings: false,
};
```



How to create an object?

There are different ways to create new objects, we are going to take a look at the 2 most common ways:

```
const person = {
  firstName: 'Emilia',
  lastName: 'Doe',
  age: 55,
  hasSiblings: false,
};
```

```
const person = {};
person.firstName = 'Kyia';
person.lastName = 'Vega';
person.age = 20;
person.eyeColor = 'blue';
```



Object Properties

Object properties can be both primitive values, other objects, and functions.

```
const person = {
  firstName: 'Emilia',
  lastName: 'Doe',
  age: 55,
  hasSiblings: false,
  mother: {
    firstName: 'Aisha',
    lastName: 'Lee',
    age: 81,
    hasSiblings: true,
  },
  fullName: function () {
    return this.firstName + ' ' + this.lastName;
  },
};
```



How to access an object property

The syntax for accessing the property of an object is:

```
objectName.property; // person.age
```

OR

```
objectName['property']; // person["age"]
```

OR

```
objectName[expression]; // x = "age"; person[x]
```



Object Methods

Whenever a property's value is a function, the property is known as a method.

```
const person = {
  firstName: 'Emilia',
  lastName: 'Doe',
  age: 55,
  hasSiblings: false,
  mother: {
    firstName: 'Aisha',
    lastName: 'Lee',
    age: 81,
    hasSiblings: true,
  },
  fullName: function () {
    return this.firstName + ' ' + this.lastName;
  },
};
```



Deleting Properties

```
const person = {
  firstName: 'Emilia',
  lastName: 'Doe',
  age: 55,
  hasSiblings: false,
};

delete person.age;
```



Exercise

< 20 minutes >

Intro: Your family owns a restaurant and you want to create a website for it. In the next slide you can find a database of your family's restaurant menu.

1. Transform the database from array of arrays to an array of objects, with a key-value pair for each of the properties.

2. Add a "method" to each of the new dishes that console logs "I am a dish of type: [typeOfDish] ". For a vegetarian dish, the function will log something like: 'I am a dish of type: vegetarian'



Exercise

< 20 minutes >

```
const menuDatabase = [
  ['Papadum', 20, 'vegetarian'],
  ['Pakora', 50, 'meat'],
  ['Tandoori Chicken', 60, 'meat'],
  ['Samosa', 50, 'vegetarian'],
  ['Butter Chicken', 139, 'meat'],
  ['Chicken Korma', 129, 'meat'],
  ['Chicken Vindaloo', 149, 'meat'],
  ['Saag Lamb', 130, 'meat'],
  ['Lam Tikka Masala', 159, 'meat'],
  ['Yellow Daal Tadka', 119, 'vegetarian'],
  ['Biryani', 129, 'vegetarian'],
  ['Gulab Jamun', 55, 'dessert'],
  ['Mango Kulfi', 35, 'dessert'],
  ['Rasmalai', 60, 'dessert'],
];
```



Break



Let’s talk about our final project!



Final Project



Final Project



Final Project



Final Project



Final Project Purpose

Create a portfolio-worthy project with all the tools and knowledge shared throughout the course so that you start accumulating coding experience for your front-end career.



Final Project Learning goals

- Reinforce and practice all the knowledge shared in the course
- Use all the concepts you have been through and put all of them together to create something beautiful
- Have your first strong project to show to your future employees
- Get support from your teachers and assistants to help you and guide you through the process so you can successfully create a project you are proud of
- Support and get support from your colleagues that have been through the same curriculum and are probably your very first colleagues.



Final Project core concepts

- HTML elements
- CSS rules and specificity
- Variables
- DOM Manipulation
- Event Listeners
- Functions
- Control flow
- Arrays
- Objects
- Loops

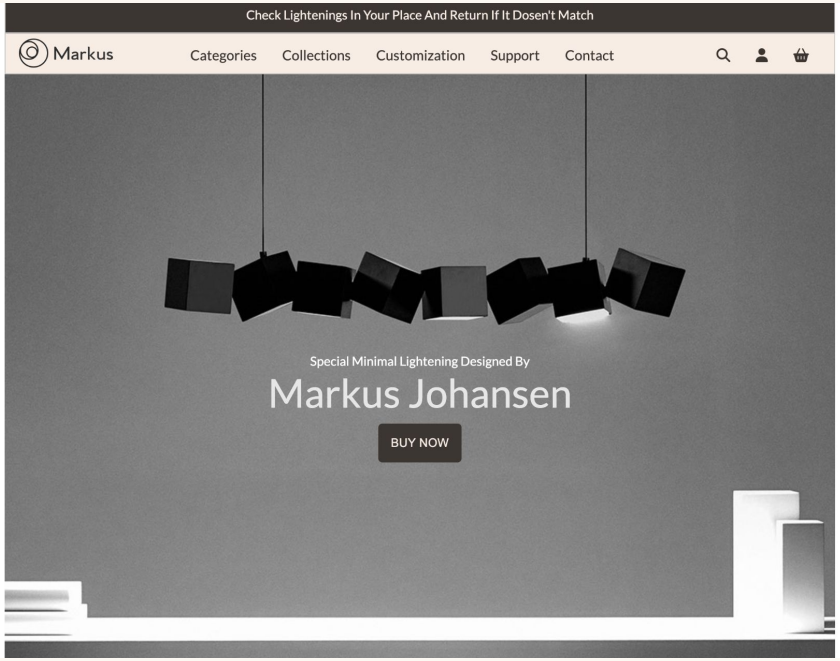


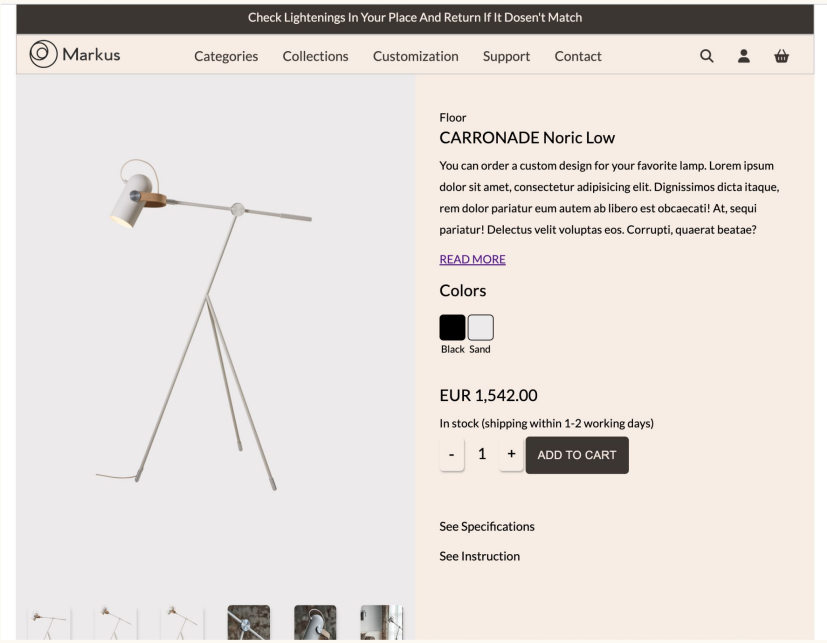
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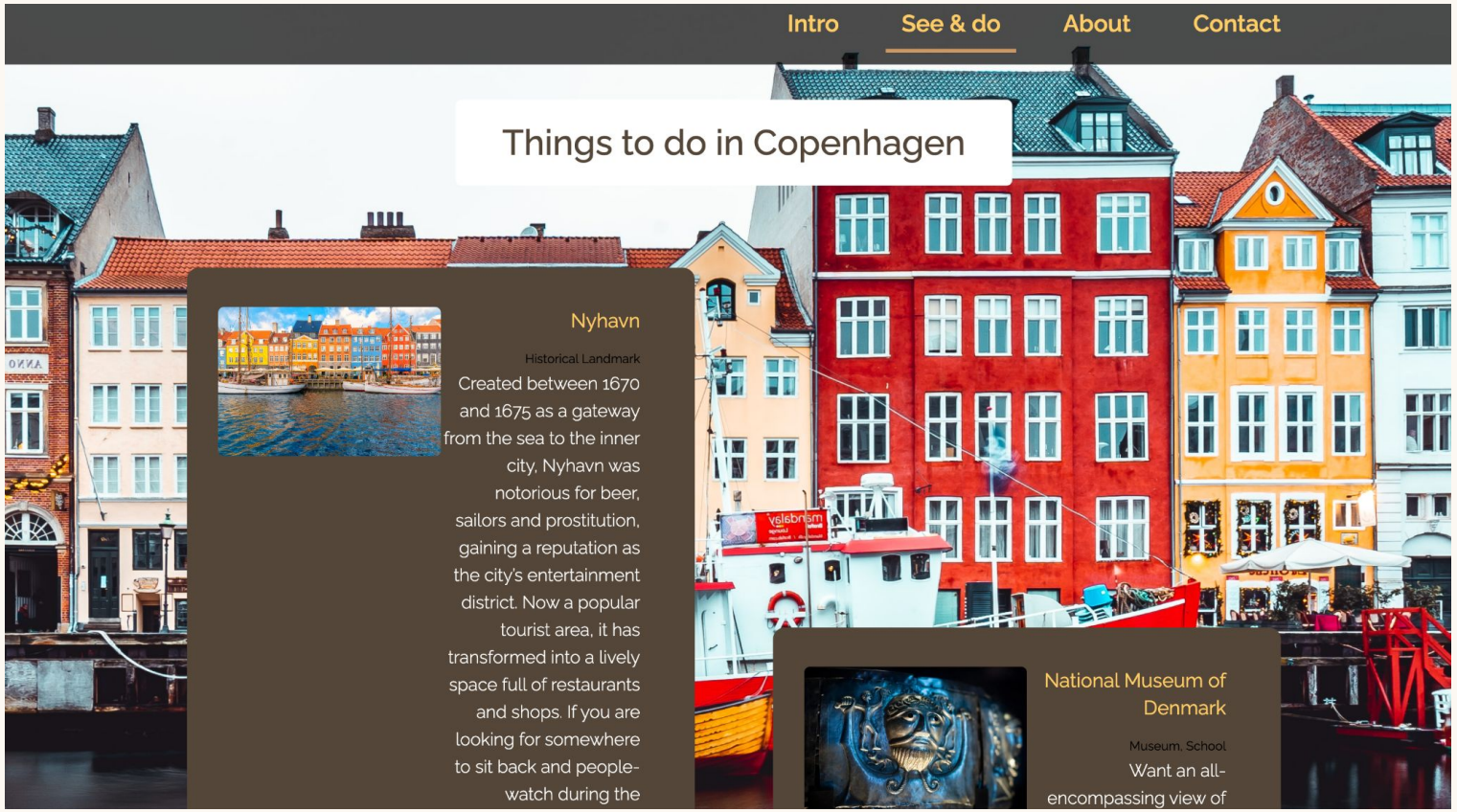


Outcome: Final Project





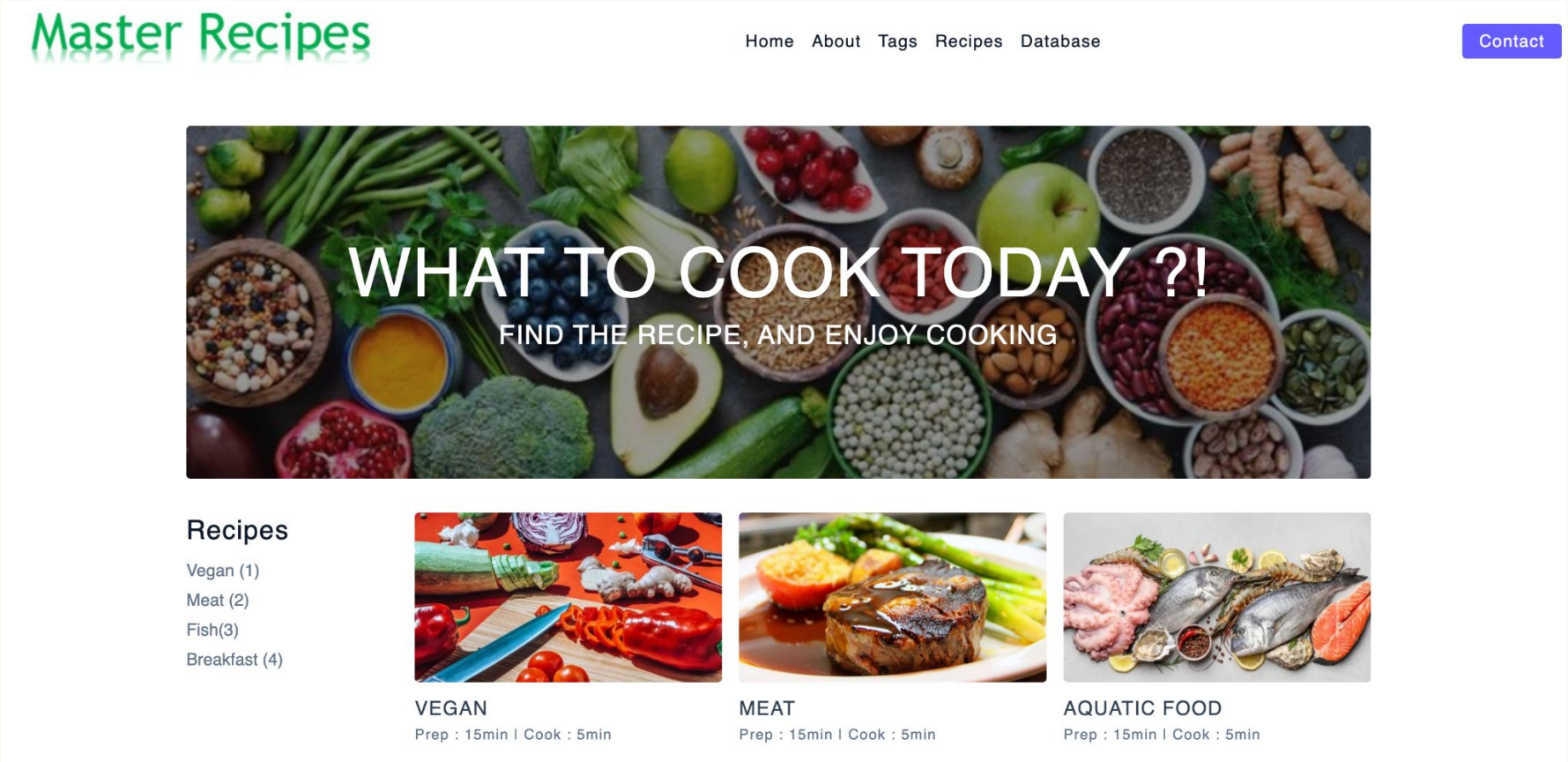
Outcome: Final Project



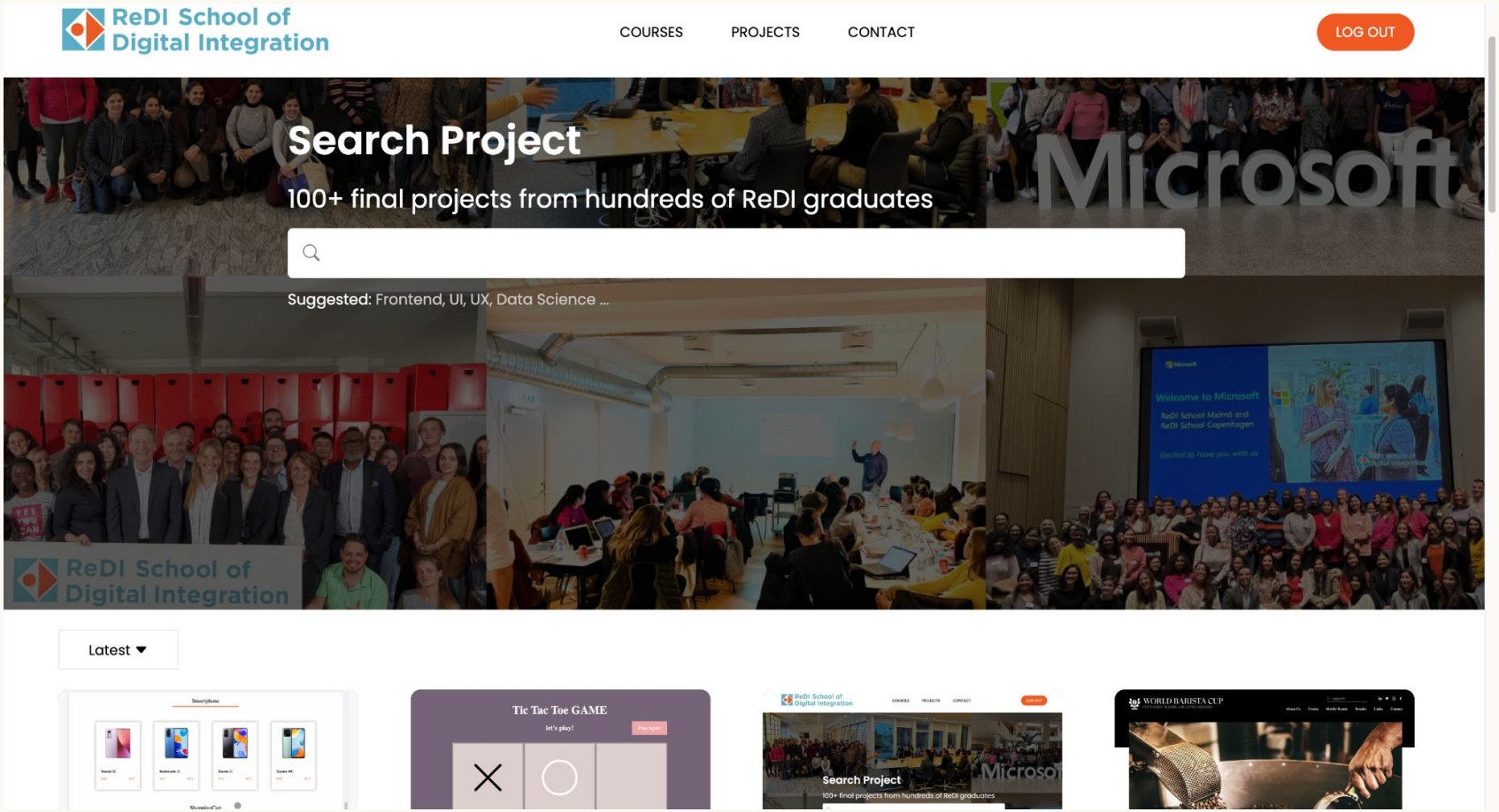
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Outcome: Final Project

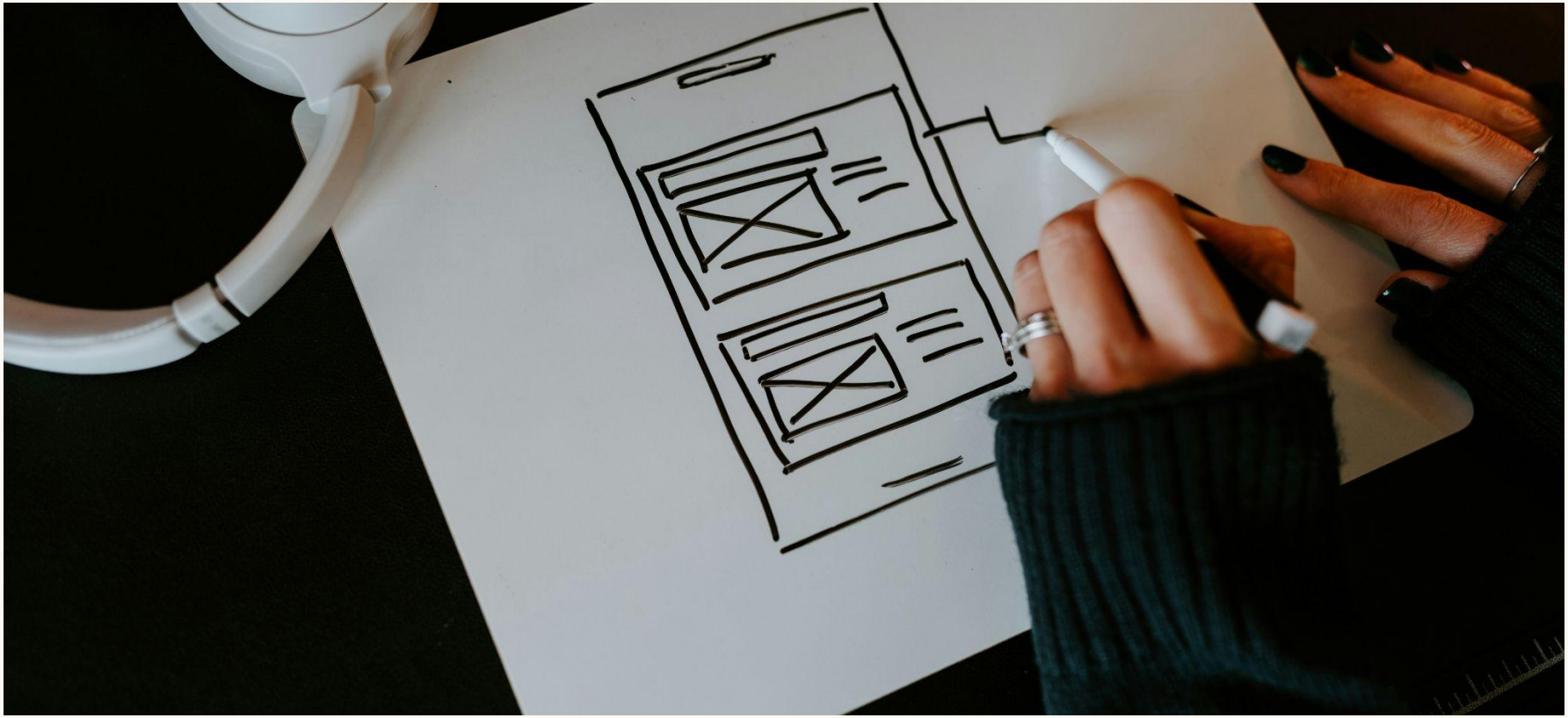


Ideation Tips for Your Final Project



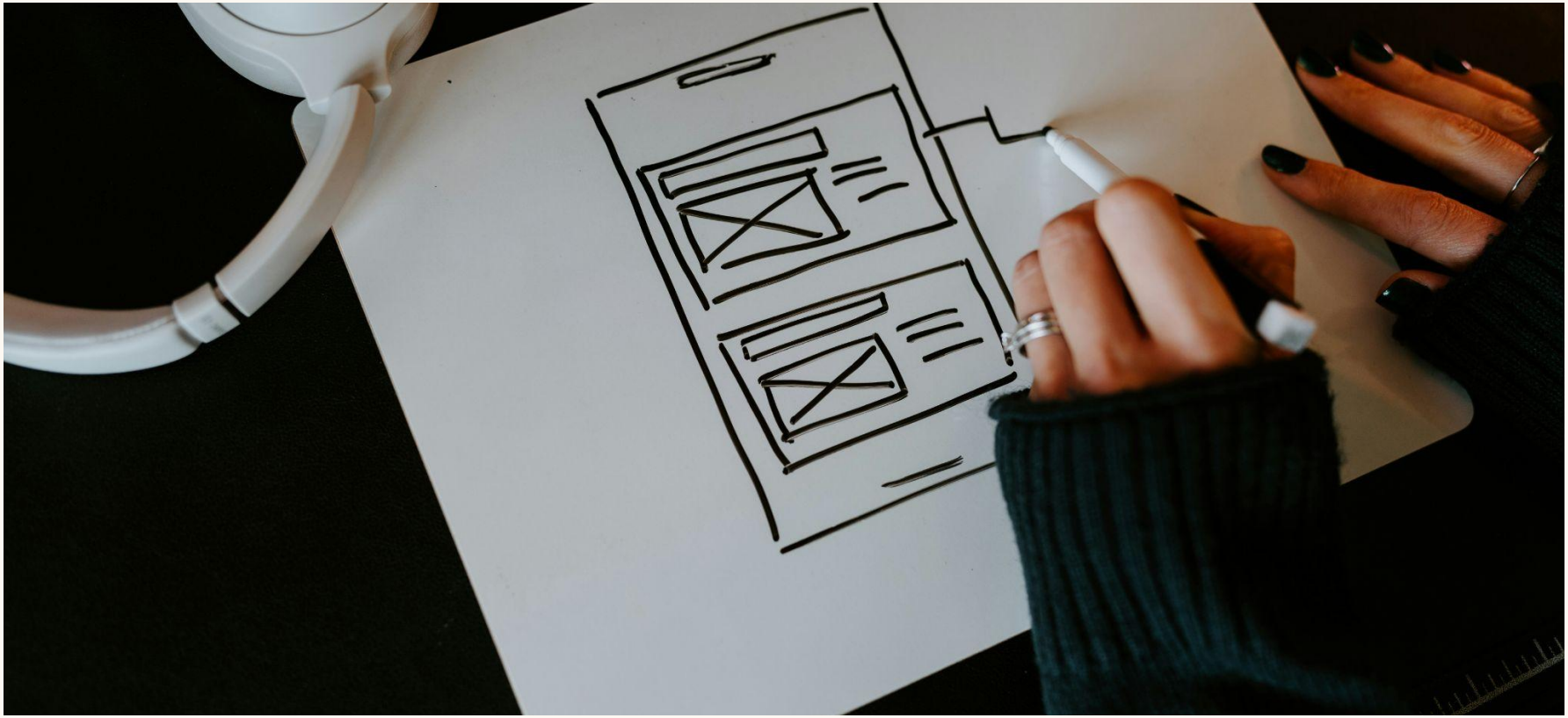
Ideation Tips - Wireframing

Wireframing is very useful during the ideation process and consists of making simple sketches or outlines of a web page or app layout. It’s like a blueprint that shows the structure, content, and main functions without any design details like colors or images.



Ideation Tips - Wireframing

Wireframes let designers and developers focus on the user experience (UX) and how information is organized before starting detailed design work and implementation.



Ideation Tips - Wireframing

Examples of tools usually used for wireframing:

- Pencil and Paper
- Miro
- Figma

More about wireframing:

- https://www.freecodecamp.org/news/what-is-a-wireframe-ux-design-tutorial-website/
- https://balsamiq.com/learn/articles/what-are-wireframes/

